# Exponential functions

Lecture and worked examples

If $a > 1$, $x \mapsto a^x$ is strictly increasing. If $0 < a < 1$, it is strictly decreasing. For $f(x) = ka^x$, multiplying by $k < 0$ reverses the direction of variation. In every case, $a^x > 0$.

In general, $a^x + a^y \neq a^{x+y}$. The rules apply to products and quotients, not to sums.

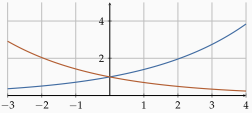
A company's turnover goes from 80,000 to 95,000 euros in four years. The overall multiplier is $Q = \frac{95{,}000}{80{,}000} = 1.1875$. The average annual multiplier is $q = 1.1875^{\frac{1}{4}} \approx 1.0440$: the average annual rate is therefore about 4.40%.

# Definition and direction of variation of $a^x$

- Make sense of $a^x$.
- Link the base to the direction of variation.
- Read curves.

# Definition and direction of variation of $a^x$

Using a calculator, complete a table of values of $1.5^x$ and $0.7^x$ for $x = -2, -1, 0, 1, 2, 3$. Plot the points, conjecture the variations and identify the point common to both curves.



# Definition and direction of variation of $a^x$

For $a > 0$, $a^x$ is defined for every real number $x$. The function is increasing if $a > 1$ and decreasing if $0 < a < 1$.

## Definition and direction of variation of $a^x$

Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

# Definition and direction of variation of $a^x$

**Running a 1-hour session:** 7 min of fluency; 10 min of sharing and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.

**Running a 2-hour session:** add 20 min of investigation, 20 min of formalising the rule, a second independent exercise and 10 min of comparative correction.

**Differentiation:** give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

# Powers and algebraic rules

- Transform powers.
- Use negative exponents.
- Spot a rule that does not apply.

# Powers and algebraic rules

Simplify $2^3 \times 2^{-5}$, $5^{x+2} \div 5^x$ and $(3^x)^2$. Then invent a numerical counterexample to $a^x + a^y = a^{x+y}$.

# Powers and algebraic rules

$a^{x+y} = a^x a^y$, $a^{x-y} = \dfrac{a^x}{a^y}$ and $(a^x)^y = a^{xy}$. These rules require the same positive base.

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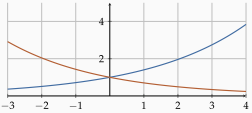
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# Functions $x \mapsto ka^x$: variation and graph

- Study $ka^x$.
- Combine sign and variation.
- Link formula and graph.

# Functions $x \mapsto ka^x$: variation and graph

Match the four functions $2^x$, $0.6^x$, $-2^x$ and $-0.6^x$ to four curves. Justify each choice using the sign, the $y$-intercept and the direction of variation.



# Functions $x \mapsto ka^x$: variation and graph

For $f(x) = ka^x$, the sign is that of $k$. If $k < 0$, multiplying by $k$ reverses the direction of variation of $a^x$.

Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

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# Average rate of change

- Calculate an average rate.
- Distinguish overall and average rates.
- Interpret an exponent $\frac{1}{n}$.

# Average rate of change

The price of a service goes from 18 to 23.50 euros in five years. Calculate the overall multiplier, then the average annual multiplier and the average annual rate. Check by applying the change five times in a row.

# Average rate of change

If $Q$ is the overall multiplier of $n$ identical changes, the average multiplier is $q = Q^{\frac{1}{n}}$ and the average rate is $q - 1$.

# Average rate of change

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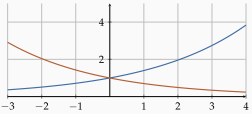
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# Reading graphical representations

- Read outputs and find the inputs that give them.
- Compare two models.
- Estimate graphically.

# Reading graphical representations

Two commercial offers are modelled by $f(x) = 120 \times 1.08^x$ and $g(x) = 180 \times 0.96^x$. Draw the curves on $[0, 12]$, read off their point of intersection and interpret the regions where each value is the greater.



# Reading graphical representations

Reading $f(x)$ means going from $x$ to the curve and then to the $y$-axis. Finding an input for a given $y$ means going from $y$ to the curve and then to the $x$-axis.

## Reading graphical representations

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# Reading graphical representations

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# Simple exponential equations

- Solve using a common base.
- Approximate a solution.
- Check plausibility.

# Simple exponential equations

Solve $4^x = 64$, $9^x = \frac{1}{3}$ and $1.12^x = 2$: first using a common base when possible, then numerically for the last one. Bracket the last solution between two integers.

# Simple exponential equations

To solve $a^x = b$: use a common base when possible; otherwise bracket the solution or use a numerical tool. Check that $b > 0$.

# Simple exponential equations

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# Simple exponential equations

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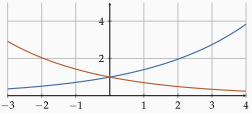
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# Modelling with an exponential function

- Build an exponential model.
- Identify the parameters.
- Predict and interpret.

# Modelling with an exponential function

An app has 12,000 users at launch and its usage grows by 9% per month. Build $f(t)$, calculate $f(6)$, then determine numerically when it will exceed 25,000 users.



# Modelling with an exponential function

A constant relative change with multiplier $q$ is modelled by $f(t) = V_0 q^t$. $V_0$ is the initial value and $q$ the multiplier per unit of time.

# Modelling with an exponential function

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# Review and chapter assessment

- Use what the chapter has covered.
- Write up a modelling solution.
- Correct a mistake.

## Review and chapter assessment

Using four pieces of student work, spot and correct: rate/multiplier confusion, a false power rule, an error in the direction of variation and a conclusion without a unit.

# Review and chapter assessment

To validate an exponential model: identify $V_0$ and $q$, calculate with the units, check the order of magnitude and interpret the result.

## Review and chapter assessment

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